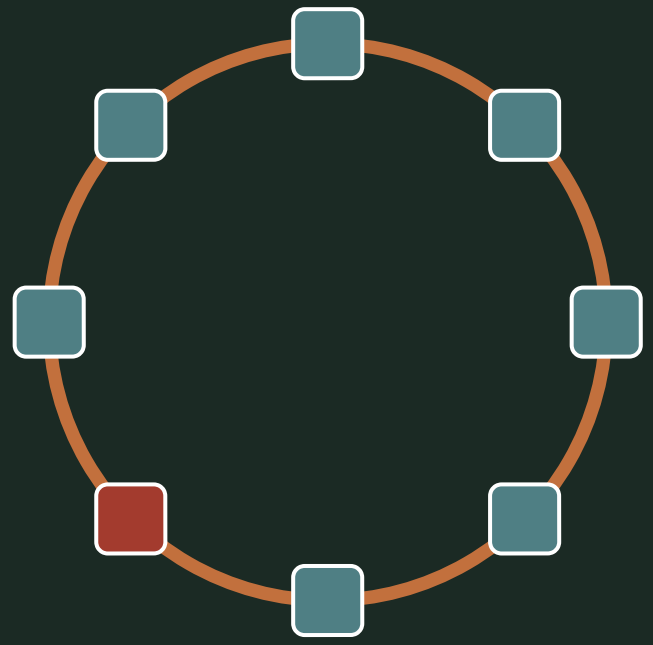
The Chip That Says No

A multi-layer hardware approach to AI safety. Each topic in two slides: the plain version with a picture, then the technical one.



The halt loop: one wire through every governor in a rack. Any one opens it and every GPU stops.

Yobie Benjamin · yobie AT ieee DOT org · linkedin.com/in/yobie

What this project is

A design for a small safety chip that sits beside every AI GPU and holds the only stamp a device needs before it acts. Built in the open as a model with tests, hardware blocks and a circuit simulation. No silicon yet, by choice.

Reference stack → **Governor design** → **FPGA, then silicon**

software, done: 37 tests, 66 attacks blocked

now: model, RTL blocks, circuit simulation, audit

only after the pre-hardware proof

Two open repositories

hardware-and-silicon: the software reference stack and the blog series. safety-governor: the chip design, model, tests, RTL, silicon flow.

One rule for the program

Nothing goes to hardware until the design is proven in software, firmware and circuit simulation. Being wrong at tape-out is the expensive mistake.

Plan of record: the rack tier

The chip rides on the server maker's management card and reaches NVIDIA's GPU board through its connector. Nothing changes in the GPU or in CUDA.

THE PICTURE

A bridge before the steel is ordered: the drawings are done and every one has been load-tested on paper. Nobody pours concrete until the drawings survive the review.

Thesis

Safety that lives only in software can be removed by software.

So the decision to act has to be made in silicon the model cannot reach, at the few places where computation turns into effect.

Every layer above the chip is a witness. Only the chip can be the lock.



Witness

a smoke alarm

- Safety training
- A judge model
- A watermark
- An observer
- A log read later



Lock

a fire door

- A ticket the device checks for itself
- A switch port that stays shut
- A response held at the exit
- A breaker that opens

THE PICTURE

A smoke alarm is a witness: it can tell you, and someone can take out the battery. A fire door is a lock: it closes whether or not anyone is listening. Most AI safety so far is smoke alarms.

The thesis, technically: a reference monitor

ANDERSON, 1972: TAMPERPROOF, ALWAYS INVOKED, SMALL ENOUGH TO VERIFY

Property	What it means here	How the governor meets it
Tamperproof	The host, the model and the operator cannot alter the decision logic or the keys	A separate die. Device secret in fuses with no read port. Boot from ROM, extend-only measurement registers, an anti-rollback counter. The rulebook changes only with k-of-n signatures from keys held outside the rack.
Always invoked	There is no path around it	A comparator at every crossing: switch ports open only for roster peers; a device acts only on a ticket; the egress holds a response until a ticket releases it; weights are written only with a ticket; power, reset and clock answer to the governor. Default deny; fail closed.
Verifiable	Small enough that its correctness can be checked	About 2 mm² of core logic in the prototype. Invariants written down: no ticket without a rule and its witnesses, no bytes to a stranger, halt on silence, no governor clears its own halt. 200 model tests, 20 RTL tests, an independent audit.

THE FIVE CROSSINGS THE CHIP STANDS AT

- **bytes leaving the GPU**
- **a request to a device that acts**
- **a response leaving the rack**
- **a write to the weights**
- **GPU power, reset and clock**

THE PICTURE

A firewall for actions, with a hand on the breaker: it decides what leaves, and when it must, it also pulls the plug.

The logic behind it

The customs officer

She cannot read the language on the paperwork. She does not need to.



the seal	the manifest	the scale	the list
unbroken?	matches the count?	matches the weight?	destination approved?

Smuggling is caught as a mismatch between what the papers say and what the scale says, not by understanding the cargo. The governor works the same way: on structure, never on meaning.

Software safety can be edited out

Refusal is a direction in the weights. Remove it and the model answers everything. An afternoon on a consumer GPU.

Watchers can only warn

A judge model is also a model: it can be fooled, patched or killed. It can hold a gate shut. It cannot be the gate.

Hardware checks structure, not meaning

A stamp, a signature, a count, a name on a list, a measured workload. A chip can refuse on those, and the model cannot forge them.

THE PICTURE

So: put the decision in silicon, and make the rulebook an allow-list. Short, signed, default no. Everything not on the list is refused by construction.

The logic, technically: what a chip can see

Seen at a crossing (small, exact, checkable)

- The attested workload: a hash of GPU firmware, host firmware and model weights, measured out of band
- The action and its typed fields: strike with a target and a munition; payments.transfer with an amount and a destination
- Counts of tokens, cycles and bytes from the GPU's own counters
- Whether the peer behind a switch port is on the roster
- The rules edition every message was signed under
- Signatures: an observer's verdict over a hash, a human's approval over a hold
- What the switches actually carried, to cross-check what was presented

Never seen

The meaning of a token. Intent. The chain of thought. Nothing in the rack can look at a stream of integers and say "that is a weapon". A design that promises to catch bad words on the wire promises what no hardware can do.

So meaning is delegated, with three guarantees

- To signed lists (protected sites, approved recipients) and to observers (classifiers, people with keys) that sign verdicts over hashes
- Always invoked: no verdict, no ticket. Cannot be impersonated: a verdict under the host's key, or the governor's own, is refused. Separate: a different key, a different box, outside the model's reach

THE PICTURE

Two bridges. Tokens and arithmetic join the software to the GPU. Hashes, signatures, counters and typed fields join the model to the governor. The chip walks only on the second.

The proposal: a referee chip beside every GPU

The model, on NVIDIA GPUs

Computes freely. Nothing it thinks has any effect until bytes leave or a request is made.

request →

The governor

Checks who is asking, whether a rule allows it, whether the witnesses agree, whether the budget holds. Then signs a ticket, or says no.

ticket →

The things that act

A weapon, a payment switch, a message gateway, the storage that holds the weights, the rack's own power. Each checks the ticket itself.



Added, not redesigned

It rides on the server maker's management card. Nothing changes in the GPU or in CUDA.



An allow-list, not a blacklist

Signed by several keys held outside the chip vendor and the operator. Everything not on it is no.



A hand on the breaker

One loop through every governor in the rack. Any one opens it, or a cut wire, and every GPU stops.



Buildable now

At the rack tier, with a server maker. The tighter tiers are asks to NVIDIA, not prerequisites.

THE PICTURE

A referee with a whistle and a rulebook he did not write: the chip holds the whistle, the rulebook is a signed document handed to it. And a breaker in the wall: it does not understand your appliances, it trips on the number.

Proposal, technically: the module and the die

Where it sits and what it reaches

- One governor per GPU slot on the OEM's management-module card (DC-SCM class)
- Per slot through the HGX connector: SMBus (SMBPBI), power brake, PERST#, PWR_EN, present and fault. HGX reset is grouped today
- NVSwitch port disable out of band through NSM; DPU steering tables through DOCA on the management port
- The GPU's confidential-computing attestation taken as an input; the BMC reads through the governor

What is on the die

- An OpenTitan-class root of trust: fuses, DICE key chain, OTBN for P-384 signatures (17 ms each, so one signature per 50 ms batch), AES-CTR plus HMAC for storage
- A mailbox of 21 signed commands, checked in order: sequence, rules edition, signature; first failure reported
- The policy image and its epoch; the gate core (lookup and comparators); the ledger (budgets, rate windows); a hash-chained log whose head lives in a register
- Meter, heartbeat, lever drivers, loop I/O and the signed message fabric between governors

Tier	Where the governor's authority comes from	Status
Rack	Management-card levers, switch and DPU management paths, the halt loop	Plan of record; needs a server OEM, nothing from NVIDIA's silicon
Board	Same firmware, policy format and fabric on a board without a baseboard connector of its own	Fallback; the gaps it leaves are recorded, not hidden
Integrated	Buffer-class checkers in the GPU's copy engines and peer MMU; per-GPU reset	Needs NVIDIA; closes the one gap the rack tier records

The layers: witnesses above, the lock below

5	**Governance root**	Signs the rulebook. Keys outside the chip vendor and the operator	*in the bank: the board that writes the rules* — LOCK
4	**Control plane: the gate**	Default deny on the only way out: rule, witnesses, budget, human hold, ticket, log	*in the bank: the teller's window* — LOCK
3	**Observers**	Fast, cheap, signed verdicts about one request; often early, sometimes wrong	*in the bank: the floor guard* — WITNESS
2	**Provenance**	Watermark and detector, signed registry; a verdict a rule can require	*in the bank: the ID check* — WITNESS
1	**Model and runtime**	The model, CUDA, the serving stack. Measured, watched, never trusted to decide	*in the bank: the customer* — UNTRUSTED
0	**Silicon root of trust**	Fused secret, anti-rollback counter, measurement registers, DICE keys, quote engine	*in the bank: the vault in the foundation* — LOCK

THE PICTURE

A bank. The board writes the rules, the teller's window enforces them, the floor guard watches for trouble, the ID is checked, the customer is served and never trusted with the keys, and the vault is poured into the foundation.

Layers, technically: mechanism, platform

Layer	Mechanism	What it signs or checks	On NVIDIA's platform
5 Governance root	k-of-n quorum over the policy image; rollback refused; expiry enforced	The rulebook (allow-list), the approved-firmware list, revocations	Keys held by a policy authority; OpenShell's policy prover checks the derived network rules
4 Control plane	Attestation check, rule lookup, witness collection, budget and rate, human hold, one-time ticket bound to a hash of the fields, hash-chained log	Tickets, log checkpoints, the compliance report	The governor module now; a DOCA application on BlueField-4 next; an I/O-stage block in the GPU eventually
3 Observers	Verdict over the hash of one request: risk, flags, expiry, under an anchored key	A signed verdict the gate requires on consequential actions	Sentry on BlueField-4 as the required witness; an independent observer as the second
2 Provenance	Watermark detector, retrieval, signed registry	A verdict {registry, digest, exact | tampered | unknown}	Runs beside the vendor's sampler
1 Model and runtime	Measured into the workload hash; routed through the switches and egress the governor controls	Nothing. It can sign nothing the gate accepts	The GPU, CUDA and the serving stack, unchanged
0 Silicon root of trust	Fused device secret, anti-rollback counter, extend-only measurement registers, DICE derivation, quote with a fresh nonce	Quotes: the measurement, the key chain, the counter	GPU confidential-computing attestation and the DPU's root, bound to the governor's own

How it meets the model

1 The model thinks

Token ids in, numbers out. No authority here and none needed.

2 The runtime asks

It turns the output into a typed request: the action and its fields.

3 The governor checks four things

Who is asking. Is there a rule. Do the witnesses agree. Is the budget intact.

4 Ticket, hold, or no

A ticket bound to the exact fields, one use, short life. Or a hold for a human. Or a refusal, logged.

5 The device checks the ticket

Against what it measures itself. A changed amount or target breaks the match.

6 Act, and record

The action happens and the record goes into a chain the chip signs.

THE PICTURE

A boarding pass. The passenger can say anything at the gate; the agent scans the pass, which names one person, one flight, one seat, and works once. The model can write "pay 40,000" all day; the ticket is the pass, and only the governor can print one.

The flow, technically: every check, every refusal

Step		What is checked	Fails with (as the model prints it)	Backed by
1	**Mailbox**	Sequence number, then rules edition, then signature; the first failure is reported	a stale or forged frame is refused before the gate sees it	H3 quorum and mailbox tests
2	**Who is asking**	The workload hash (GPU firmware, host firmware, model) and its evaluation attestation	model.not_evaluated	H1 attestation, H18 oversight
3	**Is there a rule**	Exact rule beats wildcard; typed limits; signed lists for targets and recipients	gate.constraint.target.in_denied_list, endpoint.target_protected	H6 comparator, the strike narration
4	**Do the witnesses agree**	Observer verdict from a key in the observer set, fresh, under the risk limit; human quorum on holds	gate.witness_missing, gate.held_for_human, release.hazard_class_not_allowed	H17 use cases, H18 oversight, H8 override
5	**Is the budget intact**	Compute budget and license, session hazard budget, byte budgets, rate windows	release.session_hazard_budget_exceeded, COMPUTE_BUDGET	H13 shared budget, H18
6	**The ticket**	Signed over the hash of the exact fields; one use; short expiry; bound to one endpoint	ticket.missing, ticket.replay, ticket.bad_signature, ticket.wrong_endpoint	H17: forged, replayed and wrong-hash tickets never deliver
7	**The crossings**	Switch ports open only for roster peers; the egress holds responses until released; bytes cross-checked	ep.no_capability, OUTPUT_UNPRESENTED	H2 enforcement points, the plot narration

Where it stands: drawings tested, no silicon yet

200

hypothesis tests pass in the functional model of the governor and a simulated 72-GPU rack

20

simulation tests pass on the hard-core RTL blocks, with a flow to the open 130 nm process

317 µs

worst case for the halt loop to reach the last of 72 governors, against a 1 ms budget

~2 mm²

of core logic for the governor as configured in the prototype

66

attacks blocked with the predicted reason code in the software reference stack

0

changes needed in the GPU, in CUDA or on the baseboard at the rack tier

THE PICTURE

Drawings, tested. No bricks laid. The two numbers that decide adoption, the real overhead and the false-halt rate, are still to be measured on hardware.

The challenges, stated plainly

The model
no authority needed

→

The runtime
asks for a ticket

→

The governor
full authority: signs or refuses

→

The network
partial: the door, not the conversation

→

The bank
none, unless it checks the stamp

Reach: the governor's authority ends at the last device that checks its stamp. A payment system is three hops away and has never heard of it.

Meaning

The chip makes sure an observer is asked. It cannot make the observer right about a disguised request.

False halts

The first support ticket will say: the governor halted my rack. The rate is not measured yet.

The levers

On today's HGX board, reset is grouped and the fine-grained signals sit under NDA.

Cost

One signature costs 17 ms on the crypto engine. Under one percent in the model, not yet in silicon.

Governance

Who holds the keys, whose rules govern a shared rack, and the tightest tier needs NVIDIA.

Adoption

A chip wired to nothing that acts is a logger. The first device maker to check a stamp decides this.

THE PICTURE

A firewall at the building's edge decides whether a packet leaves. It cannot decide what the server across town does with it. The fix is not to reach across town; it is to keep the key in a box the firewall controls.

Challenges, technically: mechanism, status

Challenge	What the design does	Status
Reach: a payment system two or three hops out	A: the comparator in the device. B: move the credential behind a ticket-checking adapter, so the model's runtime never holds the API key. C: open the DPU flow to the provider only under a live ticket. D: always the signed record	A and C modelled and tested (H6, H2, H17). B is a pattern the endpoint model already expresses; the three-hop payment narration is the next piece of work
Meaning: observers can be fooled	Observer always invoked, cannot be impersonated, separate from the model; hazard classes, legibility floor, session hazard budget, TRNG audit sampling with escrow	Guarantees tested (H17, H18). Observer error rates: open; needs a red team and a quorum of observers on high-hazard classes
False halts	Loop beacon inside the signed heartbeat; two-sense-period debounce; a jumpered governor detected within two heartbeats; FMEA	Modelled (H11) and simulated (ngspice). Not measured on hardware
Levers on HGX	Per-slot SMBus, power brake, PERST#, PWR_EN at the connector; grouped reset accepted as coarser; the evidence register names what is verified	Evidence register done. Design guide under NDA still needed
Cost	P-384 on OTBN at 1,697,985 cycles (17 ms); one signature per 50 ms window; engine 34 percent busy batched; firmware share 0.01 percent of GPU time	Modelled (H9). Silicon or FPGA measurement: open
Governance and adoption	k-of-n quorum with keys outside vendor and operator (G8); partitions and per-rule approvers; integrated-tier asks written	Two tenants with two quorums on one baseboard: not designed. Integrated tier: needs NVIDIA

What I need to prove or disprove the thesis

A server maker to talk to

About the management-card slot and the GPU board's design guide.

A device maker willing to check a ticket

A functional-safety vendor, or whoever runs the tools an agent calls.

A lab with a rack and an FPGA

To measure the real overhead and the false-halt rate over weeks.

A red team

To attack the release gate and the observers with disguised requests.

Chip designers and an auditor

To tear the design apart before any tape-out.

THE PICTURE

Back to the bridge: the drawings are done. What a bridge needs next is a soil report, a steel mill and an independent engineer who signs. Each of these can come back as a no, and a no is informative.

What I need, technically: who, what, and why

Who	What I need	What it settles
A server OEM	The management-module slot; the HGX design guide under NDA; the connector pinout	Whether per-module levers reach the connector. If not, the rack tier is coarser than the design says
NVIDIA	Seven firmware asks: NSM port-disable semantics, per-GPU reset, buffer-class checkers in the copy engines and peer MMU	The integrated tier; whether the recorded gap can be closed from inside the silicon
An endpoint partner	A device or tool gateway that verifies the ticket: P-384 signature, typed fields, epoch, freshness	Whether the comparator works outside the model. If nobody will check the stamp, the thesis fails in practice
A lab with an HGX or NVL rack	An FPGA governor beside a real GPU for weeks	The real overhead against the one percent goal; the false-halt rate; the loop beacon in the field
A model lab's red team	Disguised requests against the release gate and the observers	The size of the semantic residual; whether a quorum of observers is enough
Chip designers, an auditor	Review of the RTL, the firmware plan and the proof-assumption list; formal proofs of the block invariants	Verification before tape-out

What would disprove it

- No device that acts will verify a ticket
- The levers do not reach the baseboard connector
- False halts at a rate operators will not accept
- Measured overhead above one percent after batching
- Observers miss disguised requests too often for the hazard cases
- Nobody will hold the quorum keys

The one-sentence version

The governor never knows what the model is thinking and does not need to, because thinking fires nothing. A ticket does, and the model cannot mint one.

github.com/YobieBenjamin/safety-governor
github.com/YobieBenjamin/hardware-and-silicon
The Chip That Says No, parts 1 to 4, and design documents 00 to 18

Yobie Benjamin · yobie AT ieee DOT org · linkedin.com/in/yobie